Structural Change in a Multisector Model of Growth

Ngai and Pissarides (2007): Sectoral Productivity, Structural Change, and Balanced Growth

Paper presentation
Group 4
American Economic Review, 2007

Authors: L. Rachel Ngai and Christopher A. Pissarides

Group members: Binghui Guo, Heming Chen, Yuran Xia, Yunqi Gu, Zihao Liang

Reporter: Binghui Guo, Zihao Liang

Advanced Macroeconomics II

Contents

01

PART 1

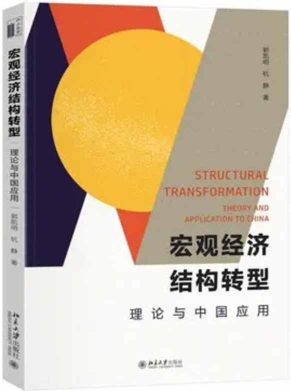
Introduction

Presenter: Binghui Guo

Course Positioning

第 1 章
价格效应与收入效应

第 2 章
要素结构与结构转型



第 3 章
需求结构与结构转型

第 4 章
技术结构与结构转型

...

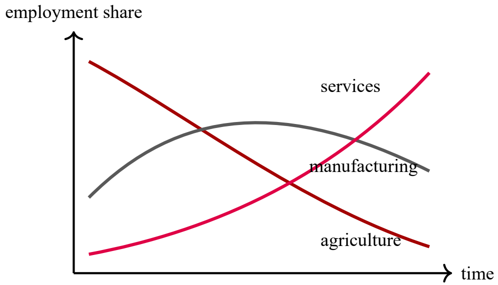
Structural Change vs. Balanced Growth

Kaldor's Stylized Facts

Labour productivity	Capital deepening
Stable return	Constant K/Y
Stable factor shares	Growth-rate differences

Theory response: Solow; Ramsey–Cass–Koopmans; endogenous growth models; sectoral proportional-growth BGP.

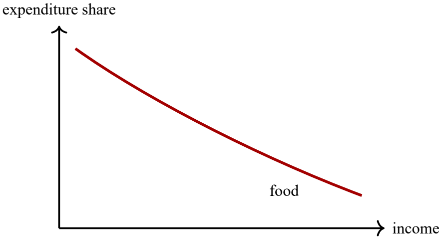
Kuznets Facts



Theory response: Baumol's cost disease; Baumol (1967); Baumol et al. (1985).

Nonhomothetic Preferences

Engel law



Engel law:

$$\eta_{\text{food}} < 1, \qquad \eta_{\text{manufacturing}}, \ \eta_{\text{services}} \geq 1.$$

- **Matsuyama (1992)**
  Stone–Geary demand + learning-by-doing $\Longrightarrow$ closed vs. open economy outcomes.
- **Echevarria (1997); Laitner (2000)**
  Income-elastic sectoral demand $\Longrightarrow$ reallocation.
- **Cost**
  Specific, sometimes *ad hoc*, preference restrictions.

Kongsamut–Rebelo–Xie (2001): Breakthrough and Cost

Stone–Geary channel

$$U = \sum_i \beta_i \log(C_i - \bar{C}_i)$$

Subsistence terms $\bar{C}_i$

↓

Income effects move sectoral demand

↓

Structural change with aggregate BGP

- **Breakthrough:** Kuznets-style reallocation can coexist with Kaldor-style aggregate ratios.
- **Cost:** subsistence demand parameters must be tied to sectoral productivity growth.
- **Problem:** preference parameters and technology parameters are no longer independent.

Roadmap of the Argument

Facts sectoral shifts → Model CES goods + TFP gaps → Mechanism prices and labor shares → Aggregate balanced growth → Extensions inputs and capital goods

Structural change

Derive how employment shares respond to relative prices and sectoral TFP growth rates.

Aggregate growth

Identify when structural change is compatible with constant aggregate ratios.

02

PART 2

Baseline Model

Presenter: Binghui Guo

Baseline Economy

$$\text{Sectors } i = 1, \ldots, m-1 \text{ consumption goods only} \longrightarrow c_i$$

$$\text{Sector } m \text{ manufacturing} \longrightarrow c_m, \ \text{capital good}$$

Labor and capital are allocated across sectors: $\{n_i, k_i\}_{i=1}^m$.

Planner's objective

$$U = \int_0^\infty e^{-\rho t} v(c_1, \ldots, c_m)\, dt.$$

Feasibility

$$\left.\begin{aligned} \sum_{i=1}^m n_i &= 1, \\ \sum_{i=1}^m n_i k_i &= k \end{aligned}\right\} \quad \text{Resource constraints}$$

$$c_i = F^i(n_i k_i, n_i), \qquad i \neq m.$$

$$\dot{k} = F^m(n_m k_m, n_m) - c_m - (\delta + \nu)k.$$

Present-Value Hamiltonian

Hamiltonian

$$
\begin{aligned}
\mathcal{H}^P = {} & e^{-\rho t} v(c_1, \dots, c_m) \\
& + q[F^m(n_m k_m, n_m) - c_m - (\delta + \nu)k] \\
& + \sum_{i=1}^{m-1} \mu_i [F^i(n_i k_i, n_i) - c_i] \\
& + \lambda \left(1 - \sum_{i=1}^{m} n_i\right) \\
& + \eta \left(k - \sum_{i=1}^{m} n_i k_i\right).
\end{aligned}
$$

Optimization logic

- $\partial \mathcal{H}^P / \partial c_i = 0$ and $\partial \mathcal{H}^P / \partial c_m = 0$: identify the shadow values of consumption goods.
- $\partial \mathcal{H}^P / \partial k_i = 0$: equalize value marginal products of capital.
- $\partial \mathcal{H}^P / \partial n_i = 0$: equalize value marginal products of labor.
- $\dot{q} = -\partial \mathcal{H}^P / \partial k$: obtain the dynamic efficiency condition.

Transversality

$$\lim_{t \to \infty} k(t) \exp\left[-\int_0^t (F_K^m(\tau) - \delta - \nu) d\tau\right] = 0.$$

Efficiency Condition

Static efficiency conditions: factor reallocation stops only when value marginal products are equal across sectors.

$$\frac{v_i}{v_m} = \frac{F_K^m}{F_K^i} = \frac{F_N^m}{F_N^i} \quad \forall i. \qquad (5)$$

Dynamic efficiency condition: the marginal utility of the manufacturing good evolves with the net return to capital.

$$-\frac{\dot{v}_m}{v_m} = F_K^m - (\delta + \rho + \nu). \qquad (6)$$

TFP Gaps & Relative Prices

Sectoral technologies

Sector i
$$F^i = A_i n_i f(k_i)$$
$$\dot{A}_i / A_i = \gamma_i$$

Sector m
$$F^m = A_m n_m f(k_m)$$
$$\dot{A}_m / A_m = \gamma_m$$

TFP gap
A_m / A_i

Same CRS technology $f(\cdot)$; sectoral paths differ only through $A_i(t)$.

- Static efficiency equalizes factor ratios: $k_i = k$.
- Relative prices carry sectoral TFP gaps.

Static efficiency

$$\frac{v_i}{v_m} = \frac{F_K^m}{F_K^i} = \frac{F_N^m}{F_N^i} \quad \forall i. \qquad (5)$$

Formula (8)

$$k_i = k, \qquad \frac{p_i}{p_m} = \frac{v_i}{v_m} = \frac{A_m}{A_i}, \quad \forall i. \qquad (8)$$

faster A_i
lower p_i
$\longrightarrow$
prices guide reallocation

CES Preferences

$$
v(c_1, \dots, c_m) = \frac{\phi(c_1, \dots, c_m)^{1-\theta} - 1}{1-\theta},
$$

$$
\phi = \left(\sum_{i=1}^{m} \omega_i c_i^{(\varepsilon-1)/\varepsilon} \right)^{\varepsilon/(\varepsilon-1)}.
$$

ε: substitution across goods

Determines whether rising relative prices increase or reduce a sector's expenditure and employment shares.

$1/\theta$: intertemporal substitution

Determines whether structural change can coexist with a balanced aggregate path.

Expenditure Ratio

- CES optimality gives $c_i/c_m = (\omega_i/\omega_m)^{\varepsilon}(p_i/p_m)^{-\varepsilon}$ **Tips:** $\left[\frac{p_i}{p_m} = \frac{v_i}{v_m}\right]$.
- Multiplying by p_i/p_m converts relative quantities into relative expenditures; substituting $p_i/p_m = A_m/A_i$ gives x_i.
- **Consumption expenditure of sector i relative to manufacturing**

$$\frac{p_i c_i}{p_m c_m} = \left(\frac{\omega_i}{\omega_m}\right)^{\varepsilon}\left(\frac{A_m}{A_i}\right)^{1-\varepsilon} \equiv x_i. \qquad (10)$$

TFP gap A_m/A_i → CES demand ε → expenditure state x_i

Aggregate Variables in Manufacturing Units

$$c \equiv \sum_{i=1}^{m} \frac{p_i}{p_m} c_i, \qquad y \equiv \sum_{i=1}^{m} \frac{p_i}{p_m} F^i.$$

Two useful simplifications

$$c = c_m X, \qquad y = A_m F(k, 1), \qquad X \equiv \sum_{i=1}^{m} x_i.$$

- x_i / X is sector i's consumption expenditure share.
- c/y is the economy-wide employment share used for consumption goods.
- $1 - c/y$ is the manufacturing labor share used to produce capital goods.

03

PART 3

Structural Change Mechanism

Presenter: Binghui Guo

Employment Shares

Consumption-only sectors

$$n_i = \frac{x_i}{X}\left(\frac{c}{y}\right), \qquad i \neq m. \qquad (13)$$

Manufacturing

$$n_m = \frac{x_m}{X}\left(\frac{c}{y}\right) + \underbrace{\left(1 - \frac{c}{y}\right)}_{\text{savings rate}}. \qquad (14)$$

- $\frac{x_i}{X}$ allocates consumption demand across final goods.
- $\frac{c}{y}$ is the labor share needed for total consumption.
- Manufacturing also supplies investment demand, captured by $1 - c/y$.

Proposition 1: Relative Employment and Relative Prices

Relative employment growth

$$\frac{\dot{n}_i}{n_i} - \frac{\dot{n}_j}{n_j} = (1-\varepsilon)(\gamma_j - \gamma_i), \qquad i,j \neq m. \qquad (15)$$

Relative price growth

$$\frac{\dot{p}_i}{p_i} - \frac{\dot{p}_j}{p_j} = \gamma_j - \gamma_i, \qquad i,j \neq m. \qquad (16)$$

Tips: $\left[\frac{p_i}{p_j} = \frac{A_j}{A_i}\right]$

Direct implication

$$\frac{\dot{n}_i}{n_i} - \frac{\dot{n}_j}{n_j} = (1-\varepsilon)\left(\frac{\dot{p}_i}{p_i} - \frac{\dot{p}_j}{p_j}\right). \qquad (17)$$

Individual Employment Dynamics

Let

$$\bar{\gamma} \equiv \sum_{i=1}^{m} \frac{x_i}{X} \gamma_i.$$

Consumption sectors: each sector follows a common intercept plus a TFP-growth term.

$$\frac{\dot{n}_i}{n_i} = \frac{\dot{(c/y)}}{c/y} + (1-\varepsilon)(\bar{\gamma} - \gamma_i), \qquad i \neq m. \tag{18}$$

Manufacturing: employment combines consumption-good production and capital-good production.

$$\frac{\dot{n}_m}{n_m} = \left[\frac{\dot{(c/y)}}{c/y} + (1-\varepsilon)(\bar{\gamma} - \gamma_m) \right] \frac{(c/y)(x_m/X)}{n_m} - \frac{\dot{(c/y)}}{1-c/y} \frac{1-c/y}{n_m}. \tag{19}$$

Proposition 2: When Do Employment Shares Change?

$$\frac{\dot{n}_i}{n_i} = \frac{\dot{(c/y)}}{c/y} + (1-\varepsilon)(\bar{\gamma} - \gamma_i). \qquad i \neq m. \qquad (18)$$

Case	Condition for structural change	Type of reallocation
Equal sectoral TFP growth	$\dot{c}/c \neq \dot{y}/y$	Between the aggregate consumption sector and the manufacturing capital-goods role.
Different sectoral TFP growth on a balanced path	$\varepsilon \neq 1$ and at least one $\gamma_i \neq \gamma_m$	Across consumption sectors with different TFP growth rates.

The Sign of $1 - \varepsilon$ Determines the Direction

$$\gamma_i < \gamma_j \quad \Longrightarrow \quad \gamma_j - \gamma_i > 0$$

$$\frac{\dot{n}_i}{n_i} - \frac{\dot{n}_j}{n_j} = (1 - \varepsilon)(\gamma_j - \gamma_i), \qquad i, j \neq m. \tag{17}$$

$\varepsilon < 1$

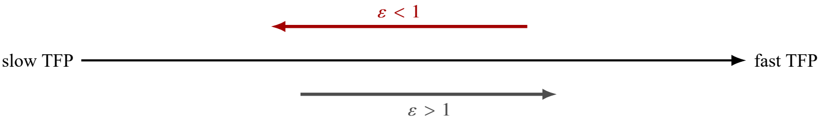
- Goods are poor substitutes.
- Demand is price inelastic.
- Labor moves toward slow-TFP-growth sectors.

$\varepsilon = 1$

- Cobb-Douglas aggregator.
- Expenditure shares are constant.
- TFP gaps change prices but not employment shares.

$\varepsilon > 1$

- Goods are strong substitutes.
- Demand is price elastic.
- Labor moves toward fast-TFP-growth sectors.



Nominal Shares Move More Than Real Shares

Nominal side

Nominal output shares:

$$\frac{p_i F^i}{p_m y} = n_i.$$

Nominal consumption shares:

$$\frac{p_i c_i}{\sum_{j=1}^{m} p_j c_j} = \frac{x_i}{X} = \frac{n_i}{c/y}.$$

Both nominal shares inherit the same relative reallocation force:

$$\frac{\dot{n}_i}{n_i} - \frac{\dot{n}_j}{n_j} = (1-\varepsilon)(\gamma_j - \gamma_i).$$

Real side

Real consumption shares:

$$\frac{\dot{c}_i}{c_i} - \frac{\dot{c}_j}{c_j} = \varepsilon(\gamma_i - \gamma_j) \qquad (20)$$

Small ε dampens changes in real consumption shares.

Mechanism Review

1. Planner problem

$$U = \int_0^\infty e^{-\rho t} v(c)\, dt$$

resources and feasibility

2. Hamiltonian

$$\mathcal{H}^P = e^{-\rho t} v + q[\cdots] + \sum_i \mu_i[\cdots] + \lambda[\cdots] + \eta[\cdots]$$

3. Optimization

$$\partial_c \mathcal{H}^P = 0, \quad \partial_{n_i,k_i} \mathcal{H}^P = 0$$
$$\dot{q} = -\partial_k \mathcal{H}^P$$

4. Efficiency conditions

$$\frac{v_i}{v_m} = \frac{F_K^m}{F_K^i} = \frac{F_N^m}{F_N^i}$$
$$-\dot{v}_m / v_m = F_K^m - (\delta + \rho + \nu)$$

5. Relative prices

$$\frac{p_i}{p_m} = \frac{v_i}{v_m} = \frac{A_m}{A_i} \quad \forall i$$

6. Expenditure state

$$\frac{p_i c_i}{p_m c_m} = \left(\frac{\omega_i}{\omega_m}\right)^{\varepsilon} \times \left(\frac{A_m}{A_i}\right)^{1-\varepsilon} \equiv x_i$$

7. Relative dynamics

$$\frac{\dot{n}_i}{n_i} - \frac{\dot{n}_j}{n_j} = (1-\varepsilon)(\gamma_j - \gamma_i)$$

8. Employment dynamics

$$\frac{\dot{n}_i}{n_i} = \frac{(c/y)^{\cdot}}{c/y} + (1-\varepsilon)(\bar{\gamma} - \gamma_i), \quad i \neq m$$

9. Nominal / real split

$$\frac{p_i F^i}{p_m y} = n_i, \quad \frac{p_i c_i}{\sum_j p_j c_j} = \frac{x_i}{X}$$

real shares move with ε

Planner $\Longrightarrow$ TFP-price state $\Longrightarrow$ expenditure state $\Longrightarrow$ employment and nominal shares

04

PART 4

Aggregate Balanced Growth

Presenter: Zihao Liang

From Structural Change to Aggregate Growth

Additional production restriction

To study a steady aggregate path, the paper specializes the common production function to Cobb-Douglas:

$$F(n_i k_i, n_i) = k_i^{\alpha} n_i, \qquad \alpha \in (0,1).$$

- Hicks-neutral technology with balanced aggregate growth requires a Cobb-Douglas form.
- The aggregate economy is measured in manufacturing-good units.

Proposition 3: Aggregate Dynamics

Given any initial $k(0)$, equilibrium aggregate dynamics satisfy:

Capital accumulation

$$\frac{\dot{k}}{k} = A_m k^{\alpha-1} - \frac{c}{k} - (\delta + \nu).$$

Euler equation

$$\theta \frac{\dot{c}}{c} = (\theta - 1)(\gamma_m - \bar{\gamma}) + \alpha A_m k^{\alpha-1} - (\delta + \rho + \nu).$$

The new term is $(\theta - 1)(\gamma_m - \bar{\gamma})$: it is zero in a one-sector model and generally time-varying under structural change.

Proposition 4: Conditions for Balanced Growth with Structural Change

Necessary and sufficient conditions

An aggregate balanced growth path with structural change exists if and only if

$$\theta = 1, \qquad \varepsilon \neq 1, \qquad \exists i \in \{1, \dots, m-1\} : \gamma_i \neq \gamma_m.$$

Why $\varepsilon \neq 1$?

Without nonunit substitution across goods, relative price changes do not generate employment-share movements on a balanced path.

Intuition

Why $\theta = 1$?

Log utility makes $\psi = 0$, so the aggregate Euler equation no longer inherits the time-varying $\bar{\gamma}$ term.

Proposition 5: Long-Run Employment Shares

Let sector l be the limiting sector: the smallest TFP-growth sector if $\varepsilon < 1$, or the largest TFP-growth sector if $\varepsilon > 1$.

Asymptotic allocation on the balanced path

$$n_m^* = \hat{\sigma} = \alpha \left(\frac{\delta + \nu + g_m}{\delta + \nu + \rho + g_m} \right), \qquad n_l^* = 1 - \hat{\sigma}.$$

- Manufacturing retains labor because it produces capital goods.
- The limiting consumption sector absorbs all consumption-goods labor.
- Other consumption sectors may decline monotonically or display hump-shaped employment shares.

05

PART 5

Extensions and Discussion

Presenter: Zihao Liang

Extension 1: Intermediate Goods

Many sectors sell output to firms as well as households. The extension lets goods be used as intermediate inputs.

Modified production function

$$F^i = A_i n_i k_i^{\alpha} q_i^{\beta}, \qquad \alpha, \beta > 0, \quad \alpha + \beta < 1.$$

Balanced-growth restriction

The intermediate-input aggregator must be Cobb-Douglas:

$$\eta = 1.$$

Under this condition, the composite productivity term A is constant over time, and a balanced growth path exists with

$$\frac{\dot{c}}{c} = \alpha A k^{(\alpha+\beta-1)/(1-\beta)} - (\delta + \rho + \nu).$$

Intermediate Goods: Employment Shares

Consumption-sector employment

$$n_i = \frac{x_i}{X}\left(\frac{c}{y}\right) + \varphi_i \beta, \qquad i \neq m.$$

Manufacturing employment

$$n_m = \left[\frac{x_m}{X}\left(\frac{c}{y}\right) + \varphi_m \beta\right] + \left(1 - \beta - \frac{c}{y}\right).$$

- The structural-change mechanism still applies to the consumption component.
- The intermediate-input term $\varphi_i \beta$ is a constant employment floor.
- In the limit, intermediate-good producers can retain employment even if their final-consumption component vanishes.

Extension 2: Many Capital Goods

Instead of a single manufacturing capital good, suppose there are κ capital-producing sectors aggregated into G.

Relative employment across capital-goods sectors

$$\frac{n_{m_j}}{n_{m_i}} = \left(\frac{\xi_{m_j}}{\xi_{m_i}}\right)^{\mu} \left(\frac{A_{m_i}}{A_{m_j}}\right)^{1-\mu}.$$

$$\frac{\dot{n}_{m_j}}{n_{m_j}} - \frac{\dot{n}_{m_i}}{n_{m_i}} = (1-\mu)(\gamma_{m_i} - \gamma_{m_j}).$$

The logic mirrors the baseline model, but the relevant substitution elasticity is now μ inside the capital-goods aggregator.

Many Capital Goods: Balanced-Growth Implication

Key result

Multiple capital-producing sectors with different TFP growth rates can coexist on an aggregate balanced growth path only if

$$\mu = 1.$$

- The aggregate capital-sector growth rate is a weighted average of capital-sector TFP growth rates.
- If $\mu \neq 1$, the weights depend on relative productivity levels and therefore change over time.
- If $\mu = 1$, the Cobb-Douglas aggregator fixes expenditure/input shares, so the aggregate growth rate is constant.

Strengths of the Paper

Theoretical strengths

- Clean separation between preferences and technologies.
- Structural change generated without nonhomothetic preferences.
- Aggregate dynamics reduce to a Ramsey benchmark under $\theta = 1$.

Interpretive strengths

- Gives a precise version of Baumol's mechanism.
- Explains why relative prices and employment can rise together.
- Reconciles large nominal shifts with small real-share shifts.

Later Literature: Income Effects and Price Effects

What later papers reassess

Ngai and Pissarides (2007) show that sectoral TFP gaps change relative prices; with poor substitutability, labor moves toward low-TFP-growth sectors.

Mechanism assessment

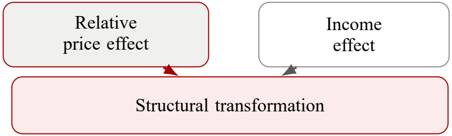
Herrendorf, Rogerson, and Valentinyi (2013, 2014) compare relative-price models with nonhomothetic-preference models: price effects fit U.S. data well, while income effects remain important.

Preference extensions

Boppart (2014); Comin, Lashkari, and Mestieri (2021); Alder, Boppart, and Müller (2022) use richer nonhomothetic preferences to capture Engel curves, income effects, and hump-shaped manufacturing employment.

Main message

Later literature does not reject Ngai and Pissarides (2007). It combines the relative-price mechanism with income effects and views structural transformation as jointly driven by both forces.



Later Literature: Other Mechanisms

Capital deepening and factor proportions

Acemoglu and Guerrieri (2008) show that capital deepening and sectoral capital-intensity differences can generate structural transformation even with equal sectoral TFP growth. **Alvarez-Cuadrado et al. (2017, 2018)** emphasize differences in capital-labor substitution.

Structural change in investment

Herrendorf, Rogerson, and Valentinyi (2021) jointly study structural change in consumption and investment, showing that investment production also changes its sectoral composition over time.

Home production and gender allocation

Moro et al. (2017) and **Ngai and Petrongolo (2017)** connect home production, rising services, female labor supply, and gender gaps to structural transformation.

Open economy and input-output linkages

Uy, Yi, and Zhang (2013) and **Sposi (2019)** show that trade, comparative advantage, and sectoral linkages can reshape a country's structural-transformation path.

Main Takeaways

1. Unequal sectoral TFP growth changes relative prices.
2. CES demand converts relative price changes into expenditure-share changes.
3. If $\varepsilon < 1$, labor moves toward sectors with lower TFP growth.
4. Aggregate balanced growth can coexist with this movement if $\theta = 1$.
5. Intermediate inputs and multiple capital goods preserve the core idea under Cobb-Douglas aggregation restrictions.

Bottom line

Structural change can be a permanent feature of sectoral allocation while the aggregate economy still looks balanced.

Q&A

Thank You

Questions and comments are welcome

06

PART 6

Appendix

Presenter: Zihao Liang

Deriving Static Efficiency: Formula (5)

The present-value Hamiltonian uses q for capital accumulation, μ_i for consumption-sector feasibility, λ for labor, and η for aggregate capital allocation.

Consumption FOCs

$$\mu_i = e^{-\rho t} v_i, \quad i < m, \qquad q = e^{-\rho t} v_m.$$

Capital-allocation FOCs

$$\mu_i F_K^i = \eta, \quad i < m, \qquad q F_K^m = \eta.$$

Combining the capital-allocation conditions gives

$$\mu_i F_K^i = q F_K^m \quad \Longrightarrow \quad \frac{\mu_i}{q} = \frac{F_K^m}{F_K^i} \quad \Longrightarrow \quad \frac{v_i}{v_m} = \frac{F_K^m}{F_K^i}.$$

The labor-allocation FOCs are

$$\mu_i F_N^i = \lambda, \qquad q F_N^m = \lambda.$$

Hence

$$\frac{v_i}{v_m} = \frac{F_N^m}{F_N^i}.$$

Putting the two margins together,

$$\boxed{\frac{v_i}{v_m} = \frac{F_K^m}{F_K^i} = \frac{F_N^m}{F_N^i} \quad \forall i}.$$

Back to Efficiency Condition

Deriving Dynamic Efficiency: Formula (6)

The aggregate capital state appears in the accumulation term and in the aggregate capital-allocation constraint. Therefore

$$\dot{q} = -\frac{\partial \mathcal{H}^P}{\partial k} = q(\delta + \nu) - \eta.$$

Using the manufacturing capital-allocation FOC, $\eta = qF_K^m$, gives

$$\dot{q} = q(\delta + \nu - F_K^m), \qquad \frac{\dot{q}}{q} = \delta + \nu - F_K^m.$$

The consumption first-order condition for the manufacturing good is $q = e^{-\rho t} v_m$, so

$$\frac{\dot{q}}{q} = \frac{\dot{v}_m}{v_m} - \rho.$$

Combining the last two equations,

$$\frac{\dot{v}_m}{v_m} - \rho = \delta + \nu - F_K^m \quad \Longrightarrow \quad \boxed{-\frac{\dot{v}_m}{v_m} = F_K^m - (\delta + \rho + \nu)}.$$

The corresponding transversality condition is

$$\lim_{t \to \infty} k(t) \exp\left[-\int_0^t (F_K^m(\tau) - \delta - \nu)\, d\tau\right] = 0.$$

Back to Efficiency Condition

Deriving Formula (8): TFP and Relative Prices

With

$$F^i(n_i k_i, n_i) = A_i n_i f(k_i),$$

the marginal products are

$$F_K^i = A_i f'(k_i), \qquad F_N^i = A_i \left[f(k_i) - k_i f'(k_i)\right].$$

Hence

$$\frac{F_N^i}{F_K^i} = \frac{f(k_i)}{f'(k_i)} - k_i \equiv \Phi(k_i), \qquad \Phi'(k) = \frac{-f(k)f''(k)}{[f'(k)]^2} > 0.$$

Formula (5) implies $F_N^i / F_K^i = F_N^m / F_K^m$. Since Φ is strictly increasing,

$$\frac{F_N^i}{F_K^i} = \frac{F_N^m}{F_K^m} \Longrightarrow \Phi(k_i) = \Phi(k_m) \Longrightarrow k_i = k_m.$$

Together with $\sum_i n_i k_i = k$, this gives $k_i = k$ for all i. Therefore

$$\frac{v_i}{v_m} = \frac{F_K^m}{F_K^i} = \frac{A_m f'(k)}{A_i f'(k)} = \frac{A_m}{A_i}.$$

Decentralized consumer optimality gives $p_i / p_m = v_i / v_m$, so

$$\boxed{k_i = k, \qquad \frac{p_i}{p_m} = \frac{v_i}{v_m} = \frac{A_m}{A_i}, \quad \forall i}.$$

Back to TFP Gaps & Prices

Deriving Expenditure Ratio: Formula (10)

CES aggregation implies

$$\phi = \left(\sum_{j=1}^{m} \omega_j c_j^{(\varepsilon-1)/\varepsilon} \right)^{\varepsilon/(\varepsilon-1)} .$$

The marginal-rate-of-substitution condition is

$$\frac{p_i}{p_m} = \frac{v_i}{v_m} = \frac{\omega_i}{\omega_m} \left(\frac{c_i}{c_m} \right)^{-1/\varepsilon} .$$

Rearranging gives relative quantities:

$$\frac{c_i}{c_m} = \left(\frac{\omega_i}{\omega_m} \right)^{\varepsilon} \left(\frac{p_i}{p_m} \right)^{-\varepsilon} .$$

Multiplying by p_i/p_m gives relative expenditures:

$$\frac{p_i c_i}{p_m c_m} = \left(\frac{\omega_i}{\omega_m} \right)^{\varepsilon} \left(\frac{p_i}{p_m} \right)^{1-\varepsilon} .$$

Using formula (8), $p_i/p_m = A_m/A_i$, yields

$$\boxed{\frac{p_i c_i}{p_m c_m} = \left(\frac{\omega_i}{\omega_m} \right)^{\varepsilon} \left(\frac{A_m}{A_i} \right)^{1-\varepsilon} \equiv x_i} .$$

Back to Expenditure Ratio

Deriving Employment Shares: Formulas (13)–(14)

Static efficiency gives $k_i = k$ and $p_i/p_m = A_m/A_i$. With $F^i = A_i n_i f(k)$,

$$\frac{p_i}{p_m} F^i = \frac{A_m}{A_i} A_i n_i f(k) = A_m n_i f(k), \qquad y = \sum_{j=1}^{m} \frac{p_j}{p_m} F^j = A_m f(k).$$

Hence

$$n_i = \frac{(p_i/p_m) F^i}{y}.$$

For $i \neq m$, $F^i = c_i$ and

$$\frac{x_i}{X} = \frac{(p_i/p_m) c_i}{c}.$$

Therefore

$$n_i = \frac{(p_i/p_m) c_i}{c} \frac{c}{y} = \boxed{\frac{x_i}{X} \frac{c}{y}}, \qquad i \neq m.$$

Manufacturing follows from $\sum_i n_i = 1$:

$$n_m = 1 - \sum_{i \neq m} \frac{x_i}{X} \frac{c}{y} = \boxed{\frac{x_m}{X} \frac{c}{y} + \left(1 - \frac{c}{y}\right)}.$$

Back to Employment Shares

Deriving Relative Employment: Formulas (15)–(17)

Start from

$$n_i = \frac{x_i}{X}\frac{c}{y}, \qquad x_i = \left(\frac{\omega_i}{\omega_m}\right)^{\varepsilon}\left(\frac{A_m}{A_i}\right)^{1-\varepsilon}.$$

For two consumption sectors $i, j \neq m$:

$$\begin{aligned}\frac{\dot{n}_i}{n_i} - \frac{\dot{n}_j}{n_j} &= \frac{\dot{x}_i}{x_i} - \frac{\dot{x}_j}{x_j} \\ &= (1-\varepsilon)(\gamma_m - \gamma_i) - (1-\varepsilon)(\gamma_m - \gamma_j) \\ &= (1-\varepsilon)(\gamma_j - \gamma_i).\end{aligned}$$

Formula (8) gives

$$\frac{p_i}{p_j} = \frac{A_j}{A_i} \quad \Longrightarrow \quad \frac{\dot{p}_i}{p_i} - \frac{\dot{p}_j}{p_j} = \gamma_j - \gamma_i.$$

Combining the two expressions yields formula (17).

Back to Proposition 1

Deriving Individual Employment Dynamics: Formula (18)

From formula (10),

$$\frac{\dot{x}_i}{x_i} = (1-\varepsilon)(\gamma_m - \gamma_i).$$

Since $X = \sum_i x_i$,

$$\frac{\dot{X}}{X} = \sum_i \frac{x_i}{X}\frac{\dot{x}_i}{x_i} = (1-\varepsilon)(\gamma_m - \bar{\gamma}), \qquad \bar{\gamma} \equiv \sum_i \frac{x_i}{X}\gamma_i.$$

For each consumption sector $i \neq m$,

$$n_i = \frac{x_i}{X}\frac{c}{y}.$$

Log-differentiate:

$$\frac{\dot{n}_i}{n_i} = \frac{\dot{x}_i}{x_i} - \frac{\dot{X}}{X} + \frac{(c\dot{/}y)}{c/y} = \boxed{\frac{(c\dot{/}y)}{c/y} + (1-\varepsilon)(\bar{\gamma} - \gamma_i)} \quad i \neq m.$$

Back to Individual Dynamics

Deriving Manufacturing Employment Dynamics: Formula (19)

Decompose manufacturing employment as

$$n_m = a_m + b_m, \qquad a_m = \frac{x_m}{X}\frac{c}{y}, \qquad b_m = 1 - \frac{c}{y}.$$

The two components grow at

$$\frac{\dot{a}_m}{a_m} = \frac{(\dot{c/y})}{c/y} - \frac{\dot{X}}{X} = \frac{(\dot{c/y})}{c/y} + (1-\varepsilon)(\bar{\gamma} - \gamma_m), \qquad \frac{\dot{b}_m}{b_m} = -\frac{(\dot{c/y})}{1 - c/y}.$$

Because the growth rate of a sum is a component-share-weighted average,

$$\frac{\dot{n}_m}{n_m} = \frac{\dot{a}_m}{a_m}\frac{a_m}{n_m} + \frac{\dot{b}_m}{b_m}\frac{b_m}{n_m}.$$

Substituting a_m and b_m gives

$$\boxed{\frac{\dot{n}_m}{n_m} = \left[\frac{(\dot{c/y})}{c/y} + (1-\varepsilon)(\bar{\gamma} - \gamma_m)\right]\frac{(c/y)(x_m/X)}{n_m} - \frac{(\dot{c/y})}{1 - c/y}\frac{1 - c/y}{n_m}}.$$

Back to Individual Dynamics

Deriving Nominal and Real Share Relations

With $k_i = k$ and $p_i/p_m = A_m/A_i$,

$$\frac{p_i F^i}{p_m y} = \frac{(A_m/A_i)A_i n_i f(k)}{A_m f(k)} = \boxed{n_i}.$$

Thus nominal output-share growth follows employment-share growth:

$$\frac{\dot{n}_i}{n_i} - \frac{\dot{n}_j}{n_j} = (1-\varepsilon)\left(\frac{\dot{p}_i}{p_i} - \frac{\dot{p}_j}{p_j}\right) = (1-\varepsilon)(\gamma_j - \gamma_i).$$

Nominal consumption shares are

$$\frac{p_i c_i}{\sum_{\ell=1}^{m} p_\ell c_\ell} = \frac{(p_i/p_m)c_i}{\sum_{\ell=1}^{m}(p_\ell/p_m)c_\ell} = \boxed{\frac{x_i}{X}}.$$

Their relative growth is the same:

$$\frac{d}{dt}\log\frac{x_i/X}{x_j/X} = \frac{\dot{x}_i}{x_i} - \frac{\dot{x}_j}{x_j} = (1-\varepsilon)(\gamma_j - \gamma_i).$$

CES demand gives

$$\frac{c_i}{c_j} = \left(\frac{\omega_i}{\omega_j}\right)^{\varepsilon}\left(\frac{p_i}{p_j}\right)^{-\varepsilon} = \left(\frac{\omega_i}{\omega_j}\right)^{\varepsilon}\left(\frac{A_i}{A_j}\right)^{\varepsilon}.$$

Therefore

$$\boxed{\frac{\dot{c}_i}{c_i} - \frac{\dot{c}_j}{c_j} = \varepsilon(\gamma_i - \gamma_j)}.$$

Back to Nominal Shares

Why $\theta = 1$ Removes the Aggregate Distortion

The aggregate Euler equation is

$$\theta \frac{\dot{c}}{c} = (\theta - 1)(\gamma_m - \bar{\gamma}) + \alpha A_m k^{\alpha - 1} - (\delta + \rho + \nu).$$

During structural change, expenditure weights x_i / X move, so

$$\bar{\gamma} = \sum_i \frac{x_i}{X} \gamma_i$$

generally changes over time.

Implication

Exact balanced growth requires the time-varying term to disappear. Setting $\theta = 1$ makes $(\theta - 1)(\gamma_m - \bar{\gamma}) = 0$.

Back